

SANSKAR SHARMA

B. Tech in Mechatronics Engineering (Pre-Final Year)

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Website: <https://sharmasanskar11.github.io/Portfolio/>

Profile & Objective

Mechatronics Engineering student with hands-on experience in embedded systems and hardware debugging, dedicated to developing innovative solutions in the field. Committed to continuous learning and collaboration, aiming to contribute effectively to cutting-edge projects that push the boundaries of technology. Eager to apply a strong foundation in engineering principles to create meaningful change in the industry.

Education

- **Puducherry Technological University**
B. Tech Mechatronics Engineering 2023-2027 (Expected) CGPA: 7.8/10
- **Trident Public School, Muzaffarpur, Bihar**
Higher Secondary Education (CBSE) (2020-2022) CGPA: 76% with Mathematics stream.
- **Sunshine Prep./ High School, Muzaffarpur, Bihar**
Secondary Education (CBSE) (2019-2020) CGPA: 86 %

Technical Skills

- Programming Languages: C++/C(Intermediate)
- Microcontrollers like Arduino UNO, MSP430, ESP32
- CNC, Lathe and Milling Machine, CAD/CAM
- Sensor Integration and Actuators
- Software like Proteus, Arduino IDE, AutoCAD

Academic Projects/Research

Prevention of Unauthorized Tampering in Electric Fence |Sep 2025 – Oct 2025 | Group Project | Smart India Hackathon, India

- Developed an IoT-based monitoring system to detect tampering in electric fences for a problem statement proposed by the Government of Kerala.
- Integrated Arduino UNO, LM393 Voltage Sensor, and ACS712 Current Sensor to continuously monitor
- Detected voltage and current fluctuations.
- Implemented GSM Module alerts to notify users instantly during unauthorized access or theft attempts.
- Contributed to hardware design, sensor calibration, and software debugging.
Technologies Used: Arduino UNO, Arduino IDE, Proteus, GSM Module

Obstacle Detection System |Mar 2024|

- Designed a distance-measurement and obstacle detection system using Ultrasonic Sensors and
- Arduino UNO.
- Displayed real-time obstacle distance on LCD screen with calibrated accuracy.

- Simulated and tested the circuit using Proteus before hardware implementation.
- Technologies Used: Arduino UNO, Proteus, Ultrasonic Sensor

Project LiEthos-India |Jan 2026- Ongoing|

- A Proposal for Ethical Energy Innovation and Decentralized Electrification.
- It aims at advancing ethical, indigenous electric vehicle (EV) battery technologies and resilient microgrid systems to address India's energy access gaps.
- Over a 0–2-year horizon, this roadmap outlines the research, development, and implementation framework for developing resilient EV batteries, vehicle-to-grid (V2G) technologies, decentralized microgrids, and curricular integration that align with national priorities.

Self-Calibrating Multi-Sensor Based Electric Motor Protection System |Jan 2026-ongoing|

- Designing and building a **low-cost adaptive motor protection system** to prevent damage due to overload, overheating, and dry-run conditions.
- Implementing **multi-sensor monitoring** using current and temperature sensors for real-time motor health assessment.
- Applied core concepts of **Mechatronics, Embedded Systems, Sensors, and Electrical Protection** in a real-world application.
- Components Used: Arduino UNO/ ESP32, ACS712 sensor, LM35 sensor, etc.

ACHIEVEMENTS & CERTIFICATIONS

- Participated in the Young Turks organised by Naukri Campus where I successfully cleared the Round 1 securing in the top 30k. Later I opted to give second round in Mechanical Engineering Paper and scored a rank of 1605 all over India.
- Forests and their Management: NPTEL (IIT KANPUR)
- Wild Life Ecology: NPTEL (IIT KANPUR)

Work Experience

Google Student Ambassador, 07/2025 to 12/2025

(Google Communique, India)

- Gained valuable public speaking experience through regular presentations to diverse audiences about the benefits of attending the university.
- Enhanced university reputation by representing the institution at various events, including open houses and college fairs.

Joint Secretary, 07/2025 to 01/2026

Puducherry Technological University AI Club – Puducherry, India (Remote)

Languages:

- English: Bilingual or Proficient (C2)
- Hindi: Intermediate (B1)
- French: A1 Beginner Level